



Aadya Arora

Final Year Undergraduate
Electrical Engineering
Minors in Artificial Intelligence
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CAREER GOAL:

My research interests span Artificial Intelligence, Machine Learning, and Computer Vision. I am specifically attracted to **AI for healthcare**, and wish to develop efficient and reliable AI systems for population scale deployment, thus contributing to saving human lives through my research contributions.

ACHIEVEMENTS

- **Direct PhD Admission:** Secured a **fully funded PhD position** at **Johns Hopkins University** (Ranked #24 globally, #1 for Medical AI) in the EECS department to specialize in Medical AI. The offer recognizes my research work and the output during the undergraduate studies at IITGN.
- **Microsoft Research & Development (Full-Time Offer):** Secured a full-time **Data Scientist role (PPO)** at **Microsoft** with highest compensation in a tech role. The offer recognizes the contributions I made during my internship improving Microsoft co-pilot for PowerPoint.
- **Microsoft Research & Development (Internship):** Secured an **off-campus internship** at **Microsoft** in a core engineering role. The offer represents one of the **most competitive selections all over India** with interview questions ranging from all computer science subjects (programming, DS, AI, ML) as well as defending my research projects.
- **Scholarly Publication & Citation Impact:** Published a single-student-author paper (with my internship supervisor) at **ICASSP 2025** (IEEE flagship conference for image and signal processing). I am currently the **only B.Tech student in the batch** with such an achievement. The paper has already received **7 citations** on Google Scholar and **100+ reads** on ResearchGate. Awarded a **25k cash prize** by IITGN recognizing my achievement.
- **International & National Research Exposure:** Only student from the B.Tech. 2026 batch to secure an **international research internship** at the **University of Bath** (Top 5 UK University) in second year of my UG studies. The internship recognized work done during my earlier research internship at **IIIT Hyderabad** (Ranked #1 in India for Computer Vision). I also followed it up with another research internship at **IISc Bangalore** (NIRF #1).
- **Professional Service (IEEE Reviewer):** Invited as an official **Reviewer for IEEE Signal Processing Letters** recognizing my published work in ICASSP. Only student in IITGN to receive such an honor. I have already completed technical peer-reviews for 4 global submissions to date in the areas of Computer Vision and Signal Processing.
- **Microsoft Research India – Academic Summit 2025:** One of the few students nationwide selected to attend India's premier annual academic summit in AI.
- **Inter-IIT Tech Meet Recognition:** Ranked in the **Top 3 for Research Paper Poster** across all 23 IITs in the EE and CSE domain; recognized for technical depth and innovation at India's most prestigious inter-collegiate technical summit.
- **Judging & Outreach:** Invited to serve as a **Technical Judge** for the Robotics competition at **Jamnabai Narsee School** alongside **Prof. Harish P.M.**, evaluating advanced engineering prototypes and fostering school-level STEM innovation.
- **Technical Stakeholder (HackRush '26):** Appointed as the **Problem Statement Setter** for the AI in Healthcare track at **IITGN's flagship hackathon**; authored the technical challenges and evaluation metrics for the competition, and evaluated the submissions.
- Featured in the **April 2026 edition of Quill Magazine**.

PUBLICATIONS

- **MedFocusCLIP: Improving Few-Shot Classification in Medical Datasets using Pixel-Wise Attention.** A. Arora, V. Namboodiri. *IEEE ICASSP*, 2025. [[ArXiv](#)]
Flagship IEEE SPS Conference, **CORE A** Ranked; h5-index: 110
- **WavShadow: Wavelet-Based Shadow Segmentation and Removal.** S. Jain*, A. Arora*, V. Vekaria, K. Gandhi, S. Raman. *ICVGIP*, 2024. (*Equal Contribution) [[ArXiv](#)]
Premier Indian Conference in Computer Vision; h5-index: 18
- **TEMPEST: A Machine Learning Framework for Battery Thermal Prediction and Chemistry Selection.** A. Arora*, S. Patil*, P. Bhardwaj. *IEEE Transactions on Industry Applications (TIA)*, 2026.. (*Equal Contribution)[[IEEEExplore](#)]
Q1 Journal; Impact Factor: 4.4; h5-index: 78
- **Battery ETHOS: Electro-Thermal Model for State Estimation.** S. Patil*, A. Arora*, P. Bhardwaj. Under Review at *IEEE COMPEL*, 2026. (*Equal Contribution) [[GitHub](#)]
Q2 Journal; Impact Factor: 1.5; h5-index: 25

INTERNSHIPS

• Microsoft

May 2024 – July 2024

Data Scientist Intern (Received Full-Time PPO)

[Certificate](#)

- Worked with the **Copilot team for PowerPoint**, focusing on how AI can be integrated into PPT to assist in creating better and more structured presentations.
- Built datasets and trained models by extracting and structuring **text and formatting information from slides**, where content is represented in JSON format (including attributes like bold, italics, layout, etc.).
- Developed data pipelines to generate diverse formatting examples, helping models better understand slide structure and improve automated formatting suggestions.
- Iteratively improved model performance through error analysis and refinements, leading to a **Pre-Placement Offer (PPO)**.

• VAL Lab, Indian Institute of Science (IISc)

Dec. 2024 – Apr. 2025

Research Intern (Advisor: Prof. R. Venkatesh Babu)

[Demo](#)

- Worked on understanding how diffusion models, especially Diffusion Transformers (DiTs), behave internally, with the goal of designing more efficient image editing methods that require less compute and time.
- Performed layer-wise analysis to study how features evolve across the network, which helped in identifying where different attributes like structure and style are captured.
- Explored attention mechanisms and designed unified attention masks to reduce prompt leakage in multi-region text-to-image generation, improving control over specific parts of the image.
- Ran LoRA and DreamBooth experiments across different transformer blocks to understand how editing at different depths affects style retention and visual consistency.

• University of Bath

May 2024 – July 2024

Research Intern (Advisor: Prof. Vinay Namboodiri)

[GitHub](#)

- Worked on AdaptHIPIE alongside my exploration of MedFocusCLIP, initially aiming to extend similar ideas to the medical domain, but the lack of annotated data led me to rethink the approach toward more data-efficient methods.
- Developed AdaptHIPIE by introducing lightweight adapter modules into the HIPIE architecture, showing that strong performance in open-vocabulary referring segmentation can be achieved with relatively low additional compute.
- Achieved an mIoU of 85.15 on the RefCOCO dataset, improving over prior methods through better visual-text alignment using CLIP features.
- Analyzed failure cases in hierarchical segmentation, especially the distinction between instance-level and part-level understanding, and used these insights to refine how adapters influence different levels of representation.

• CVIT Lab, IIIT Hyderabad

Dec. 2023 – Jan. 2024

Research Intern (Advisors: Prof. CV Jawahar, Prof. Vineeth N. B.)

[Demo](#)

- My first research project, where I explored failure cases in autonomous driving, particularly in unstructured traffic environments using the Indian Driving Dataset (IDD).
- Focused on identifying corner cases, such as rare or unusual objects and scenarios that standard detection models often miss.
- Experimented with class-agnostic Region Proposal Networks (RPNs) to capture these uncommon instances beyond predefined categories.
- Built a pipeline using semantic segmentation and common-class suppression to surface safety-critical anomalies like wrong-way driving.

SELECTED PROJECTS

• Training-Free Fine-Grained 3D Scene Editing

Jan. 2026 – Present

Advisor – Prof. Shanmuganathan Raman, IIT Gandhinagar

[Demo](#)

- Worked on enabling fine-grained 3D scene editing without requiring per-scene training, by combining diffusion-based editing with 3D Gaussian representations.
- Used attention-based control in diffusion models to manipulate object-level properties such as size and shape, while preserving identity and visual consistency.
- Designed a mechanism to map 2D edits from diffusion outputs into a consistent 3D space across multiple views, ensuring coherence in reconstructed scenes.
- Built an efficient pipeline for generating full 360-degree edited scenes in seconds, reducing common issues such as view inconsistency and texture artifacts.

• Thermal-Vision Interpretability

Aug. 2025 – Nov. 2025

Advisor – Prof. Nipun Batra, IIT Gandhinagar

[Demo](#)

- Started by analyzing results from the ThermEval benchmark, which showed that Vision-Language Models often fail on thermal reasoning tasks despite strong performance on RGB data.
- Investigated this by working with open-source VLMs and generating attention heatmaps and patch-wise similarity maps to understand how models process thermal inputs.
- Identified a consistent misalignment in CLIP-based models, where attention and similarity scores prioritized background regions over semantically meaningful thermal objects.
- Addressed this by implementing attention prompting strategies to guide models toward object-relevant regions, improving reasoning and grounding on thermal imagery.

• Caffeine Consumption and Cognitive Performance

Aug. 2025 – Nov. 2025

Advisor - Prof. Vaibhav Tripathi, Integrative Neuroscience & Technology Lab

[Demo](#)

- Designed and conducted a behavioral study to understand how varying caffeine levels affect attention, reaction time, and cognitive control using the Flanker Task.
- Investigated cognitive phenomena such as the speed-accuracy trade-off and placebo effects, observing how physiological and psychological factors influence decision-making.
- Collected and controlled real-world data across participants, accounting for confounding variables such as sleep patterns, study habits, and prior caffeine intake.
- Applied statistical analysis (ANOVA, t-tests, non-parametric tests) to evaluate group differences and uncover underlying cognitive trends.

TEACHING EXPERIENCE

• Academic Discussion Hours (ADH) Mentor

Jan. 2026 – Present

Principles of Electrical Engineering (ES 116)

IIT Gandhinagar

- Appointed as an **ADH Mentor** for a large-scale introductory course with over 300 first-year students.
- Conducted weekly doubt-clearing sessions to support the diverse learning needs of the undergraduate cohort.

• Teaching Assistant

Jan. 2025 – Apr. 2025

Probability, Statistics & Data Visualization (ES 114)

IIT Gandhinagar

- Served as a **Teaching Assistant** for a foundational course with 400+ students, focusing on data-driven reasoning and visualization techniques.
- Engineered specialized **coding notebooks** and instructional materials to facilitate hands-on learning in Python, graded assessments and mock quizzes.

• Teaching Assistant

Aug. 2024 – Nov. 2024

Machine Learning (ES 335)

IIT Gandhinagar

- Supported a class of 250+ students by developing assignments and teaching materials focused on core **Machine Learning** algorithms.
- Led technical tutorials and coding sessions, graded assignments and exams.
- Delivered guest lectures, helped in building demo notebooks.

POSITIONS OF RESPONSIBILITY

- **Management Coordinator**, PDC, IIT Gandhinagar *Aug. 2025 – Apr. 2026*
 - Led the planning and execution of institute-wide career development initiatives at IIT Gandhinagar, coordinating outreach, logistics, and student engagement across multiple cohorts as part of the Professional Development Council.
- **Coordinator-Career Exposure and Guidance**, PDC, IIT Gandhinagar *Aug. 2024 – Apr. 2025*
 - Coordinated mentorship programs, internship preparation sessions, and career guidance initiatives for undergraduate students across disciplines through the Professional Development Council at IIT Gandhinagar.
- **Student Guide**, Student Support Services, IIT Gandhinagar *Aug. 2023 – Apr. 2024*
 - Mentored first-year students on academic, extracurricular, and personal matters as part of a faculty-supported guidance team.
- **General Member**, PDC, IIT Gandhinagar *Aug. 2023 – Apr. 2024*
 - Assisted in organizing professional development events and outreach activities aimed at enhancing student readiness for internships and higher studies.